# Msc Thesis Topic

**“Norwegian Sentiment Classification on NoReC: Fine-Tuning vs Prompting with Open-Source Language Models”**

Supervisor: [To be agreed]

# Background & Motivation

Sentiment classification is a central NLP task with clear practical value for analysing opinions in large-scale text collections. For Norwegian, fine-tuned transformer-based models such as NB-BERT can achieve strong performance, while simpler classical pipelines remain attractive due to speed and low computational requirements. At the same time, open-source instruction-tuned language models enable zero-shot and few-shot classification through prompting, potentially reducing the need for task-specific training. This thesis will provide a controlled and reproducible comparison between (i) classical baselines, (ii) fine-tuned NB-BERT, and (iii) prompted open-source language models, with particular focus on whether small language models can match larger models and/or fine-tuned approaches for Norwegian sentiment classification.

## Objectives

- Establish classical baseline performance for Norwegian sentiment classification on NoReC.
- Fine-tune NB-BERT for the same task under a controlled experimental setup.
- Evaluate a limited set of open-source instruction-tuned language models in zero-shot and few-shot settings using a fixed prompting protocol.
- Compare small vs large language models and contrast prompted approaches with fine-tuning and classical baselines.
- Conduct an error analysis to identify systematic weaknesses and common misclassifications across model families.

## Methodology

1. **Data and Task Definition**
   The study will use the NoReC (Norwegian Review Corpus) and define a sentiment classification task based on review ratings. A reproducible label mapping and dataset split strategy will be defined and documented.

2. **Baseline Model Development (Classical)**
   Supervised text classifiers based on TF–IDF features will be developed as strong baselines (e.g., Logistic Regression and Linear SVM). Model selection will be performed using a validation strategy within a controlled setup.

3. **Fine-Tuned Transformer Baseline (NB-BERT)**
   NB-BERT will be fine-tuned for sequence classification using standard transformer fine-tuning procedures. Training configuration and model selection will be based on validation performance.

4. **Prompt-Based LLM Evaluation (Zero-shot and Few-shot)**
   A limited set of open-source instruction-tuned language models will be evaluated using a fixed prompting protocol in zero-shot and few-shot settings. The selection will include both small and larger parameter sizes to enable a controlled small-vs-large comparison.

5. **Evaluation and Error Analysis**
   All approaches will be evaluated on a held-out test set using standard classification metrics (including Accuracy and Macro-F1). Comparisons will include performance and computational cost where relevant. An error analysis will be conducted using quantitative summaries (e.g., confusion patterns) and qualitative inspection of typical misclassifications, optionally including analyses by review length and/or domain when relevant metadata is available.

6. **Expected Deliverables**

   - Thesis report describing dataset, methodology, experiments, results, and discussion
   - Reproducible implementation (code, configuration, documentation)
   - Final results summary comparing classical baselines, fine-tuned NB-BERT, and zero-/few-shot prompted LLM approaches, including small-vs-large findings and key error patterns